Lecture 6: Ensembling and Entity Embedding

Deep Learning for Actuarial Modeling | Summer School 'Lifelong Learning' | Università Cattolica del Sacro Cuore, Milano

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ABSTRACT

The first part of this lecture presents network ensembling, called nagging. The second part considers covariate engineering which is an important part of regression modeling. This involves dealing with categorical covariates, but it also deals with bringing continuous covariates into a suitable form.

1 Introduction

OVERVIEW

- The 1st part of this lecture presents *network ensembling*, also called *nagging*.
- Network ensembling is a methodology to reduce randomness in SGD fitted networks and to improve predictive performance.
- The 2nd part covers covariate engineering.
- Covariate engineering makes up a major part of regression modeling in actuarial problems.
  - First, *categorical covariates* need to be brought into a numerical form. Usually, this is done by an *embedding*.
  - Second, *continuous variables* may need a suitable transformation, or they can be embedded as well.

This lecture covers parts of Chapters 2 and 5 of Wüthrich *et al.* (2025).

2 Network ensembling

- A critical point of network fitting is that it involves *several elements of randomness*. Even for a fixed architecture and fitting procedure, one typically has *infinitely many equally good* fitted models ('solutions') on a finite learning sample.
- The elements of randomness involve:
  1. the initialization of the network weight $\vartheta^{[0]}$ for SGD;
  2. the random partition into learning sample $\mathcal{L}$ and test sample $\mathcal{T}$;
  3. the random partition into training sample $\mathcal{U}$ and validation sample $\mathcal{V}$;

4. the random partition into the mini-batches $(\mathcal{U}_k)_{k=1}^{\lfloor n/s \rfloor}$.
5. There are further random items like drop-outs, etc.

- This makes early stopped SGD solutions (highly) non-unique. This non-uniqueness is typical for machine learning solutions, and it is troublesome in actuarial work.

2.1 Ensembling/nagging

- Breiman (1996) introduced *bagging* for regression trees.
- Bagging combines **b**ootstrap and **agg**regat**ing**. Bootstrap is a re-sampling technique, and this is combined with aggregation which has an averaging effect, reducing the randomness.
- We replace the bootstrap by different SGD 'solutions', because network fitting has the above mentioned items of randomness, we naturally receive multiple solutions.
- Ensembling of network predictors was introduced by Dietterich (2000a) and Dietterich (2000b). Subsequently, it was studied in an actuarial context by Richman and Wüthrich (2020), where it was called *nagging* for **n**etwork **agg**regat**ing**.

2.2 Ensemble predictor

- Having multiple (conditionally) i.i.d. predictors $(\widehat{\mu}_j)_{j=1}^M$, one builds the *ensemble predictor*

$$\widehat{\mu}^{(M)} = \frac{1}{M} \sum_{j=1}^M \widehat{\mu}_j.$$

- This ensemble predictor has an *estimation uncertainty*

$$\sqrt{\mathrm{Var}\left(\widehat{\mu}^{(M)}\right)} = \frac{1}{\sqrt{M}} \sqrt{\mathrm{Var}(\widehat{\mu}_1)} \ \to \ 0 \qquad \text{for } M \to \infty.$$

- The important takeaway is that ensembling over (conditionally) i.i.d. predictors substantially reduces estimation uncertainty.

- Caveat: This does not say anything about a bias of the estimated model for the true model!

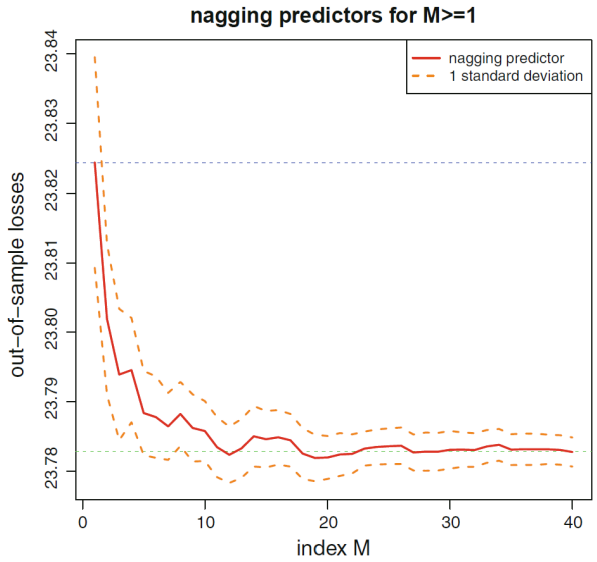
2.3 Nagging predictor

- Based on learning sample $\mathcal{L} = (Y_i, \boldsymbol{X}_i, v_i)_{i=1}^n$, choose M conditionally i.i.d. fitted FNNs, where the conditionally i.i.d. applies to the elements of randomness in SGD fitting; see above.
- This gives us M conditionally i.i.d. FNNs $(\mu_{\widehat{\vartheta}_j})_{j=1}^M$, given $\mathcal{L}$.
- This motivates the *nagging predictor*

$$\widehat{\mu}_M^{\text{nagg}}(\boldsymbol{X}) = \frac{1}{M} \sum_{j=1}^M \mu_{\widehat{\vartheta}_j}(\boldsymbol{X}).$$

- This ensembling reduces the fluctuations from SGD fitting by a factor $\sqrt{M}$.
- The next plot shows over how many FNNs we need to ensemble to arrive at an optimal forecast model w.r.t. out-of-sample loss.

Out-of-sample Poisson deviance losses as a function of $M \geq 1$.



- This is the French MTPL claims count example.
- We conclude that we need roughly 10 to 20 i.i.d. fitted FNNs.

(The y-scale uses the number of policies instead of the total exposure.)

2.4 Results: nagging predictor

model	*in-sample loss*	*out-of-sample loss*	*balance (in %)*
Poisson null model	47.722	47.967	7.36
Poisson GLM	45.585	45.435	7.36
Poisson FNN	44.846	44.925	7.17
nagging predictor	44.849	44.874	7.36

- For the nagging predictor we used $M = 10$ individual network fits.
- As expected, we receive an out-of-sample improvement.

RECOMMENDATION

In network predictions, always consider the nagging predictor $\widehat{\mu}_M^{\text{nagg}}$ with $M \in \{10, 20\}$. This likely improves the forecast model.

3 Categorical covariates

- Categorical covariates (nominal or ordinal) need pre-processing to bring them into a numerical form.
- Categorical covariate pre-processing is done by *entity embedding*.

- Consider a categorical covariate X taking values in a finite set $\mathcal{A} = \{a_1, \ldots, a_K\}$ having K different *levels*.
- The running example in this lecture has $K = 6$ levels:

$$\mathcal{A} = \Big\{ \text{accountant},\ \text{actuary},\ \text{economist},\ \text{quant},\ \text{statistician},\ \text{underwriter} \Big\}.$$

- We need to bring this into a numerical form for regression modeling.

3.1 Ordinal categorical covariates

For *ordinal* (*ordered*) levels $(a_k)_{k=1}^K$, use a 1-dimensional entity embedding

$$X \in \mathcal{A} \ \mapsto\ \sum_{k=1}^K k\, \mathbf{1}_{\{X = a_k\}} \in \mathbb{R}.$$

Our running example has an alphabetical ordering.

level	***embedding***
accountant	1
actuary	2
economist	3
quant	4
statistician	5
underwriter	6

One may argue that the alphabetical order is risk insensitive (not useful).

3.2 One-hot encoding

One-hot encoding maps each level a_k to a basis vector in $\mathbb{R}^K$

$$X \in \mathcal{A} \;\mapsto\; \left(\mathbf{1}_{\{X=a_1\}}, \ldots, \mathbf{1}_{\{X=a_K\}}\right)^\top \in \mathbb{R}^K.$$

level						
accountant	1	0	0	0	0	0
actuary	0	1	0	0	0	0
economist	0	0	1	0	0	0
quant	0	0	0	1	0	0
statistician	0	0	0	0	1	0
underwriter	0	0	0	0	0	1

One-hot encoding does not lead to full rank encodings because there is a redundancy (the first $K-1$ components are sufficient).

3.3 Dummy coding

Dummy coding selects a reference level, e.g., $a_2 = \text{actuary}$. Based on this selection, all other levels are measured relative to this reference level

$$X \in \mathcal{A} \;\mapsto\; \left(\mathbf{1}_{\{X=a_1\}}, \mathbf{1}_{\{X=a_3\}}, \mathbf{1}_{\{x=a_4\}}, \ldots, \mathbf{1}_{\{X=a_K\}}\right)^\top \in \mathbb{R}^{K-1}.$$

level					
accountant	1	0	0	0	0
actuary	0	0	0	0	0
economist	0	1	0	0	0
quant	0	0	1	0	0
statistician	0	0	0	1	0
underwriter	0	0	0	0	1

This leads to full rank, but also to sparsity on large datasets, i.e., most of the entries are zero (for K large). This may be problematic numerically.

3.4 Entity embedding

- Borrowing ideas from natural language processing (NLP), use low dimensional *entity embeddings*, with proximity related to similarity; see Brébisson *et al.* (2015), Guo and Berkhahn (2016), Richman (2021).
- Choose an *embedding dimension* $b \in \mathbb{N}$ – this is a hyper-parameter selected by the modeler, typically $b \ll K$.
- An *entity embedding* (EE) is defined by

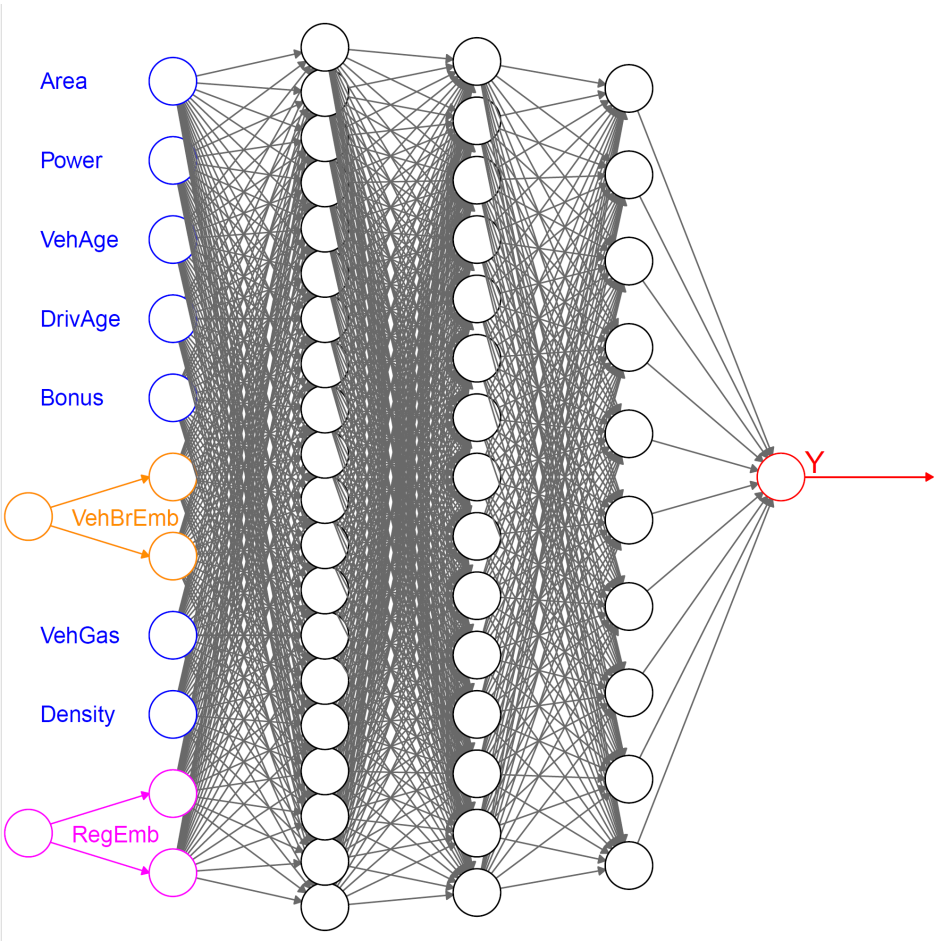
$$\boldsymbol{e}^{\mathrm{EE}} : \mathcal{A} \to \mathbb{R}^b, \qquad X \mapsto \boldsymbol{e}^{\mathrm{EE}}(X).$$

- In total this entity embedding involves $b \cdot K$ *embedding weights*.
- The embedding weights need to be determined either by the modeler (manually) or during the model fitting procedure (algorithmically), and proximity in embedding should reflect similarity in (risk) behavior.

- Manually chosen example with $b \cdot K = 24$ embedding weights.

level	*finance*	*maths*	*stats*	*liability*
accountant	0.5	0	0	0
actuary	0.5	0.3	0.5	0.5
economist	0.5	0.2	0.5	0
quant	0.7	0.3	0.3	0
statistician	0	0.5	0.8	0
underwriter	0	0.1	0.1	0.8

- Note: the weights do not need to be normalized, i.e., only proximity in $\mathbb{R}^b$ is relevant.
- In neural networks, this can be achieved by setting up a so-called *embedding layer*. The embedding weights can be learned with stochastic gradient descent – an embedding layer can be seen as an additional FNN layer which is *not fully-connected*.



3.5 Target encoding

- Especially for regression trees, one uses *target encoding*.
- Assume a given sample $(Y_i, X_i, v_i)_{i=1}^n$ with categorical covariates $X_i \in \mathcal{A}$, real-valued responses Y_i and weights $v_i > 0$.
- Compute the weighted sample means on all levels $a_k \in \mathcal{A}$ by

$$\overline{y}_k = \frac{\sum_{i=1}^n v_i Y_i \, \mathbf{1}_{\{X_i = a_k\}}}{\sum_{i=1}^n v_i \mathbf{1}_{\{X_i = a_k\}}}.$$

- These weighted sample means $(\overline{y}_k)_{k=1}^K$ are used like ordinal levels

$$X \in \mathcal{A} \;\mapsto\; \sum_{k=1}^K \overline{y}_k \, \mathbf{1}_{\{X = a_k\}}.$$

- Though convincing initially, this does not consider interactions between covariates, potentially giving misleading results for scarce levels. Moreover, it may suffer from a *leakage of information*.

3.6 Credibilitized target embedding

- For scarce levels, one should *credibilitize* target encoding; see Micci-Barreca (2001) and Bühlmann (1967).
- Goal: Assess how credible the individual estimates $\overline{y}_k$ are.
- Improve unreliable ones by mixing them with the global weighted empirical mean $\overline{y} = \sum_{i=1}^n v_i Y_i / \sum_{i=1}^n v_i$, providing

$$\overline{y}_k^{\text{cred}} = \alpha_k \, \overline{y}_k + (1 - \alpha_k) \overline{y},$$

  with *credibility weights* for $1 \leq k \leq K$

$$\alpha_k = \frac{\sum_{i=1}^n v_i \mathbf{1}_{\{X_i = a_k\}}}{\sum_{i=1}^n v_i \mathbf{1}_{\{X_i = a_k\}} + \tau} \;\in\; (0, 1].$$

- *Shrinkage parameter* $\tau \geq 0$ is a hyper-parameter that needs to be selected by the modeler.

4 Continuous covariates

- In theory, continuous covariates do not need any pre-processing.
- In practice, it might be that continuous covariates do not provide the right functional form, or they may live on the wrong scale.
- E.g., we may replace a positive covariate $X > 0$ by a 4-dimensional pre-processed covariate

$$X \mapsto (X, \log(X), \exp\{X\}, (X-10)^2)^\top.$$

- In GLMs, one often discretizes continuous covariates by binning, e.g., by building age classes. For a finite partition $(I_k)_{k=1}^K$ of the support of the continuous covariate X, assign categorical labels $a_k \in \mathcal{A}$ to X by setting

$$X \mapsto \sum_{k=1}^K a_k \mathbf{1}_{\{X \in I_k\}}.$$

4.1 Standardization and MinMaxScaler

- Gradient descent fitting methods require that all covariate components live on the same scale – otherwise early stopping may not be successful.
- Assume to have n instances with a continuous covariate $(X_i)_{i=1}^n$.
- *Standardization* considers the transformation

$$X \mapsto \frac{X - \widehat{m}}{\hat{s}},$$

with empirical mean $\widehat{m}$ and empirical standard deviation $\hat{s} > 0$ of the sample $(X_i)_{i=1}^n$.

- The *MinMaxScaler* is given by the transformation

$$X \mapsto 2 \frac{X - \min_{1 \le i \le n} X_i}{\max_{1 \le i \le n} X_i - \min_{1 \le i \le n} X_i} - 1.$$

- The scaling constants need to be stored!

4.2 Continuous covariate embedding

- In recent deep learning architectures, multi-dimensional entity embedding has been successfully applied to continuous covariates $X \in \mathbb{R}$, too.
- This allows one to bring continuous covariates into the same b-dimensional embedding space as an entity embedded categorical covariates.
- Select a deep FNN $\boldsymbol{z}^{(d:1)} : \mathbb{R} \to \mathbb{R}^b$ giving the entity embedding

$$X \;\mapsto \boldsymbol{z}^{(d:1)}(X) \in \mathbb{R}^b.$$

- The parameters of this embedding are learned during SGD model fitting.

- We apply entity embedding to the entire covariate vector

$$\boldsymbol{X} = (X_1, \ldots, X_{q_c}, X_{q_c+1}, \ldots, X_q)^\top,$$

  with q_c categorical covariates and $q - q_c$ continuous covariates.
- Applying entity embedding to both the continuous covariates and the categorical covariates with the same embedding dimension b, gives us an *input tensor*

$$\left[\boldsymbol{e}_1^{\mathrm{EE}}(X_1), \ldots, \boldsymbol{e}_{q_c}^{\mathrm{EE}}(X_{q_c}), \boldsymbol{z}_{q_c+1}^{(d:1)}(X_{q_c+1}), \ldots, \boldsymbol{z}_q^{(d:1)}(X_q)\right] \;\in\; \mathbb{R}^{b \times q}.$$

- The input data $\boldsymbol{X}$ is now in a tensor structure, suitable to enter *attention layers* and the *Transformer architecture* of Vaswani *et al.* (2017); see also Richman, Scognamiglio and Wüthrich (2025).

5 Example: French MTPL

data – entity embedding

```
FNN.embed <- function(seed, qq, kk, bb){
    k_clear_session(); set.seed(seed); set_random_seed(seed)
    #
    Design   <- layer_input(shape = c(qq[1]), dtype = 'float32')
    Volume   <- layer_input(shape = c(1), dtype = 'float32')
    #
    # define the entity embeddings (levels are encoded by integers)
    BrandEmb = Design[,qq[1]-1] %>%
      layer_embedding(input_dim = kk[1], output_dim = bb[1], input_length
        = 1) %>% layer_flatten()
    #
    RegionEmb = Design[,qq[1]] %>%
      layer_embedding(input_dim = kk[2], output_dim = bb[2], input_length
        = 1) %>% layer_flatten()
    #
--- to be continued ---
```

```
--- continued ---
    #
    # concatenate entity embeddings and remaining covariates for the FNN
    Network = list(Design[,1:(qq[1]-2)], BrandEmb, RegionEmb) %>%
      layer_concatenate() %>%
      layer_dense(units=qq[2], activation='tanh') %>%
      layer_dense(units=qq[3], activation='tanh') %>%
      layer_dense(units=qq[4], activation='tanh') %>%
      layer_dense(units=1, activation='exponential')
    #
    Output = list(Network, Volume) %>% layer_multiply()
    #
    keras_model(inputs = c(Design, Volume), outputs = c(Output))
    }
```

- Levels are encoded by integers.
- After embedding into $\mathbb{R}^b$, they are concatenated with the (standardized) continuous covariates before entering the feature extractor.

```
...
_________________________________________________________________
_______
 Layer (type)           Output Shape              Param   Connected to
==================================================================
=======
 embedding (Embedding)  (None, 2)                 22
['tf.__operators__.getit
 embedding_1 (Embeddin  (None, 2)                 44
['tf.__operators__.getit
                                                          'flatten_1[0]
[0]']
 dense_3 (Dense)        (None, 20)                240     ['concatenate[0]
[0]']
 dense_2 (Dense)        (None, 15)                315     ['dense_3[0][0]']
 dense_1 (Dense)        (None, 10)                160     ['dense_2[0][0]']
 dense (Dense)          (None, 1)                 11      ['dense_1[0][0]']
                                                          'input_2[0][0]']
==================================================================
=======
Total params: 792 (3.09 KB)
Trainable params: 792 (3.09 KB)
Non-trainable params: 0 (0.00 Byte)
...
```

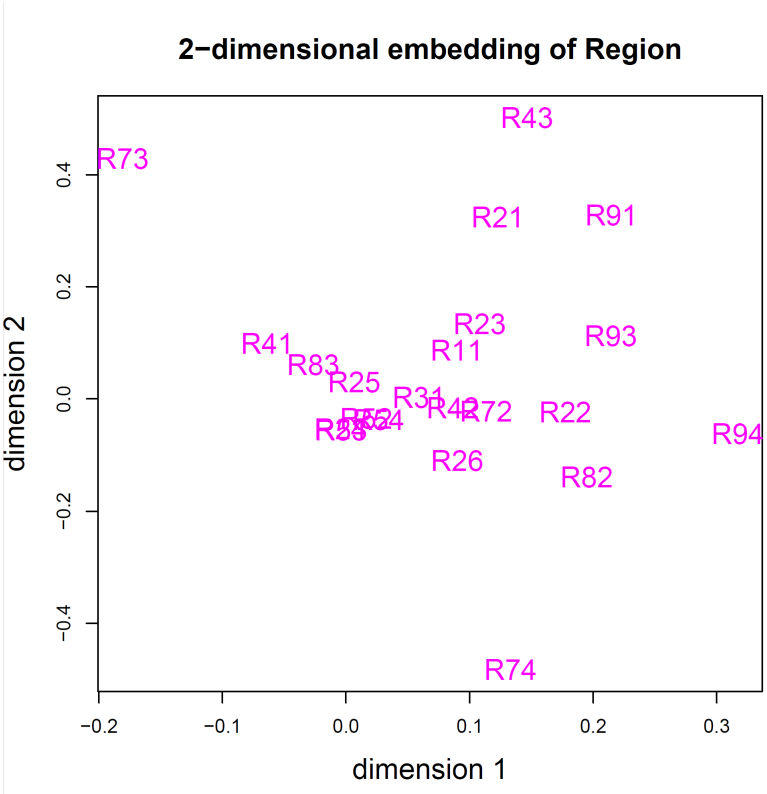
Results

model	*in-sample loss*	*out-of-sample loss*	*balance (in %)*
Poisson null model	47.722	47.967	7.36
Poisson GLM	45.585	45.435	7.36
Poisson FNN	44.846	44.925	7.17
nagging predictor	44.849	**44.874**	7.36
entity embedding FNN	45.030	44.948	7.16
embed nagging predictor	44.636	**44.864**	7.36

- For the nagging predictor we used $M = 10$ individual network fits.

- We receive slightly better results using entity embedding over one-hot encoding. This is a general observation.
- In the case of high-cardinality categorical features, one should also regularize, see Richman and Wüthrich (2024).

Illustration of 2D embeddings



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- This notebook and these slides are part of the project “AI Tools for Actuaries”. The lecture notes can be downloaded from:

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5162304

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